

EDA Report - DecodeLabs Internship

Project 2 - Exploratory Data Analysis

Date: 2026-05-15

STEP 1: Basic Statistics

Average Order Value (TotalPrice): 1053.97

Median Order Value: 823.62

Minimum Order: 11.39 | Maximum Order: 3456.40

Insight: Mean > Median = Right Skewed Data. High value orders pulling average up.

STEP 2: Trends

Top Product: Printer (181 orders)

Lowest Product: Phone (156 orders) - Needs marketing attention!

Top Payment Method: Online (258 transactions)

Top Referral Source: Instagram (259 customers)

RED FLAG: Cancelled(250) + Returned(247) = 497 problematic orders = 41% of all orders!

Business Action: Investigate cancellation and return reasons urgently.

STEP 3: Outlier Detection

Method Used: IQR (Interquartile Range)

Normal Range: 0 to 3330.41

Total Outliers Found: 8 orders above 3330.41

Highest Outlier: ORD200789 - Tablet - 3456.40

Business Action: Flag for investigation - VIP bulk orders or data entry errors?

STEP 4: Correlation Analysis

UnitPrice vs TotalPrice: 0.72 (STRONG) - Price drives revenue!

Quantity vs TotalPrice: 0.62 (MODERATE) - Bulk orders = Higher bills

Quantity vs ItemsInCart: 0.65 - Bulk buyers have more items in cart

UnitPrice vs ItemsInCart: 0.00 - Premium buyers buy less but spend more!

FINAL BUSINESS RECOMMENDATIONS

1. Investigate 41% cancellation/return rate - biggest business risk!
2. Invest more in Instagram marketing - highest traffic source!
3. Promote Phone category - lowest sales need a boost!
4. Focus on UnitPrice optimization - strongest revenue driver!
5. Investigate 8 outlier orders for fraud or VIP treatment!